The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 September 1992

INCH-POUND

MIL-S-19500/384C
3 January 1992
SUPERSEDING
MIL-S-19500/384B
6 October 1972

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON HIGH-POWER
TYPES 2N3584, 2N3585, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, high power transistors. Two levels of product assurance are provided for this device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-66).

1.3 Maximum ratings. T_C = +25°C, unless otherwise specified.

Type	P_T T_A = +25°C 1/	P_T T_C = +25°C 2/	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$	V_{CER}
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W	V dc
2N3584	2.5	35	375	250	6.0	1.0	2.0	-65 to +200	5.0	300
2N3585	2.5	35	500	300	6.0	1.0	2.0	-65 to +200	5.0	400

1/ Derate linearly, 14.85 mW/°C for T_A > +25°C.
2/ Derate linearly, 200 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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1.4 Primary electrical characteristics.

Limits	h_{FE} 1/		C_{obo}	$\|h_{fe}\|$	$V_{BE(sat)}$ 1/	$V_{BE(sat)}$ 1/	Switching	
							t_{on}	t_{off}
	I_C = 1 A dc V_{CE} = 10 V dc	I_C = 100 mA dc V_{CE} = 10 V dc	100 KH < f < 1 MHz V_{CB} = 10 V dc I_E = 0	f = 5 MHz I_C = 200 mA dc V_{CE} = 10 V dc	I_C = 1.0 A dc I_B = 0.1 A dc	I_C = 1 A dc I_B = 125 mA dc	(See table I and figure 2 herein)	
			pF		V dc	V dc	µs	µs
Minimum	25	40		3.0				
Maximum	100		120	15	1.4	0.75	3.0	3.0

1/ Pulsed.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

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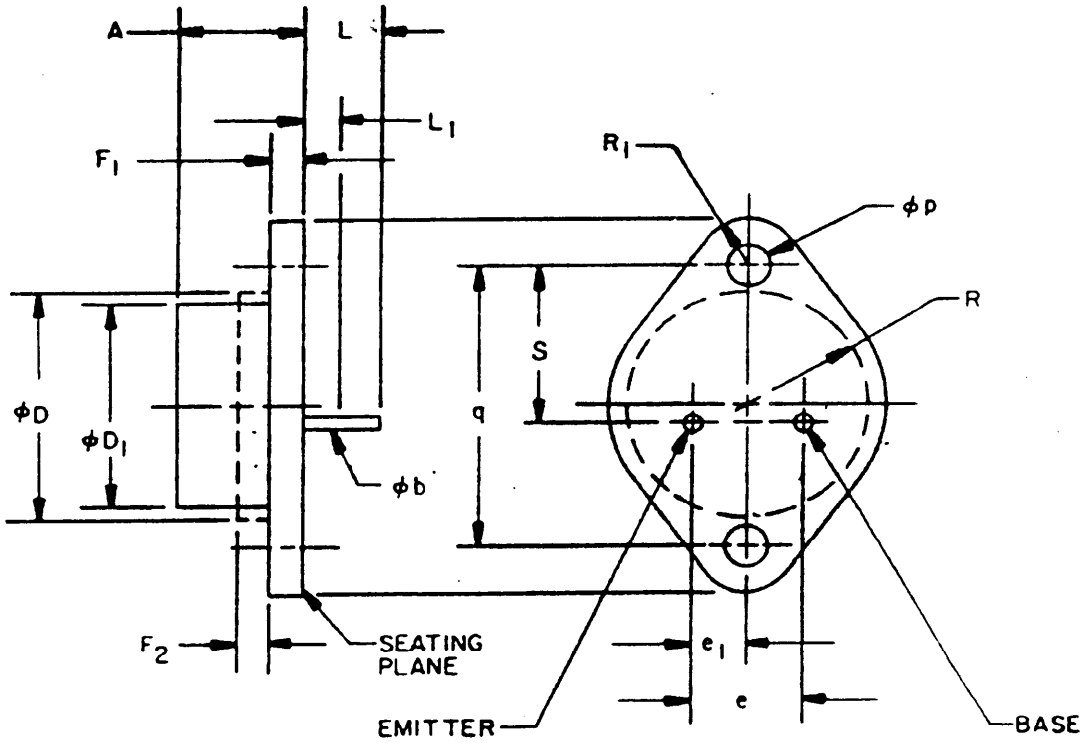


FIGURE 1. Physical dimensions (TO-66).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.340	6.35	8.64	
ϕb	.028	.034	0.71	0.86	5,9
ϕD	---	.620	---	15.75	3
ϕD_1	.470	.500	11.94	12.70	3
e	.190	.210	4.83	5.33	4
e_1	.093	.107	2.36	2.72	4
F_1	.050	.075	1.27	1.91	3
F_2	---	.050	---	1.27	3
L	.360	.500	9.14	12.70	5
L_1	---	.050	---	1.27	5,9
ϕ_p	.142	.152	3.61	3.86	7
q	.958	.962	24.33	24.43	
R	---	.350	---	8.89	
R_1	.115	.145	2.92	3.68	6
S	.570	.590	14.48	14.99	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Body contour is optional within zone defined by ϕD, ϕD_1, E_2.
4. These dimensions should be measured at points .050 (1.27 mm) to .055 (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. ϕb applies between L_1 and L. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions (TO-66) - Continued.

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following markings may be omitted from the body of the device:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.3 Screening (JANTXV and JANTX levels). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels
1/	Thermal response (see 4.3.2)
9	
11	I_{CEX1} and h_{FE2}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or .1 mA dc; Δh_{FE} = ±20% of initial value.

1/ This test shall be performed anytime before screen 9.

4.3.1 Power burn-in. Power burn-in conditions are as follows:

P_T = 2.5 W
V_{CB} = 150 V dc
T_A = +25°C, minimum

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

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4.3.2 Thermal response (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The ΔV_{BE} conditions (I_H and V_H) and maximum limit V_{BE} shall be derived by each vendor. The chosen ΔV_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted, read, and measurements recorded. The chosen ΔV_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. I_M measurement -10 mA.

b. V_{CE} measurement voltage - - - - - - - - - - - - - - - - - - -10 V (same as V_H).

c. I_H collector heating current - - - - - - - - - - - - - - - - -2 A (minimum).

d. V_H collector-emitter heating voltage - - - - - - - - - - - - -10 V (minimum).

e. t_H heating time -100 ms.

f. t_{MD} measurement delay time - - - - - - - - - - - - - - - - -50 µs to 80 µs.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IV (JANTX and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500, and table IV herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.5 Methods of inspection. Methods of inspection and test shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131, MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.5 A dc.

b. Collector to emitter voltage magnitude shall be 15 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 5°C/W.

4.5.3 SB_{IC} test. Set V_{CC} = 100 V dc without pulse applied. Adjust generator to apply a single pulse of 1 second duration to obtain I_C = 350 mA dc. The 1 second pulse width shall not be exceeded in adjustment and not more than one pulse shall be applied in any 20 second interval. (See figure 2.)

4.5.4 SB_{ICEX} test. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 2.0 A dc. Device fails test if second breakdown occurs as indicated by collapse of oscilloscope display. (See figure 3.)

4.5.5 SB_{ICER} test. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 1.4 A dc. Device fails test if second breakdown occurs as indicated by collapse of oscilloscope display. (See figure 3.)

4.5.6 V_{BR}(CEO) and V_{BR}(CER) tests. The breakdown voltages BV_{CEO} and BV_{CER} shall not be measured on a curve tracer. BV_{CEO} and BV_{CER} should be measured by means of the test circuit shown on figure 3. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 200 mA. Device fails test if collector to emitter voltage is less than breakdown voltage limit at 200 mA.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Pulsed (see 4.5.1), I_C = 10 mA dc; bias condition D	$V_{(BR)CEO}$			V dc
2N3584				250		
2N3585				300		
Breakdown voltage, collector to base	3001	Bias condition D, I_C = 15 mA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$			V dc
2N3584				375		
2N3585				500		
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6.0 V dc	I_{EBO}		0.5	mA dc
Collector to emmiter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N3584		V_{CE} = 300 V dc			1.0	mA dc
2N3585		V_{CE} = 400 V dc			1.0	mA dc
Collector-emitter cutoff current	3041	Bias condition D, V_{CE} = 150 V dc	I_{CEO}		5.0	mA dc
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 1 A dc, I_B = 0.1 A dc	$V_{BE(SAT)}$		1.4	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 1 A dc, I_B = 0.125 A dc	$V_{CE(sat)}$		0.75	V dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE1}	25	100	
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	h_{FE2}	40		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current 2N3584 2N3585	3041	Bias condition A, V_{BE} = -1.5 V dc V_{CE} = 300 V dc V_{CE} = 400 V dc	I_{CEX2}		 2.0 2.0	 mA dc mA dc
Subgroup 4						
Second breakdown, collector current, base forward biased		V_{CE} = 100 V dc (see figure 2 and 4.5.3)	SBIc	350		mA dc
Pulse response transfer ratio	3251	Test condition A, except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 30 V dc (see figure 2), I_C = 1 A dc, I_B = 100 mA dc, R_C = 29 ohms	t_{on}		3.0	µs
Turn-off time		V_{CC} = 30 V dc (see figure 2), I_C = 1 A dc, I_{B1} = -I_{B2}, = 100 mA dc, R_C = 29 ohms	t_{off}		7.0	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 200 mA dc, f = 5 MHz	$\|h_{fe}\|$	3.0	15	
Open circuit (output capacitance)	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		120	pF
Small-signal short circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 1 A dc, f = 1 kHz	h_{fe}	25	200	

1/ See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C, power application time = 1 second, 1 cycle				
Test 1		V_{CE} = 17.5 V dc, I_C = 2 A dc				
Test 2		V_{CE} = 100 V dc, I_C = 350 mA dc				
Test 3						
2N3584		V_{CE} = 250 V dc, I_C = 37 mA dc				
2N3585		V_{CE} = 300 V dc, I_C = 17 mA dc				
Second breakdown, collector current, base reverse biased		L = 100 µH, R_{BE} = 20 ohms, V_{BE} = 4 V dc (see figure 3 and 4.5.4)	SBI_{CEX}	2.0		A dc
Second breakdown, collector current, base with resistance return		L = 500 µH, R_{BE} = 50 ohms, (see figure 3 and 4.5.5)	SBI_{CER}	1.4		A dc

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group B inspection for JANTX devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1 2/		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	Test condition C
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table V, steps 1, 2, and 4.
Subgroup 3		
Intermittent operation life	1037	ΔT_J = +100°C, 2,000 cycles.
Electrical measurements		See table V, steps 1, 2, 6, and 7.
Subgroup 4		
Decap-internal visual (design verification)	2075	
Bond strength	2037	Test condition A. All internal leads for each device shall be pulled separately.
Subgroup 5		
Not applicable		
Subgroup 6		
High temperature life (nonoperating)	1031	T_{STG} = +200°C.
Electrical measurements		See table V, steps 1, 2, and 4.
Subgroup 7		
Not applicable		

1/ For sampling plan, see MIL-S-19500.
2/ A separate sample may be pulled for each test.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1.
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition B.
Terminal strength (tension)	2036	Test condition A, weight = 10 pounds, time = 15 s.
Hermetic seal	1071	
Fine leak Gross leak		
Moisture resistance	1021	
Electrical measurements		See table V, steps 1, 2, and 4.
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	10 G.
Constant acceleration	2006	
Electrical measurements		See table V, steps 1, 2, and 3.
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Intermittent operation life	1037	ΔT_J = +100°C, 6,000 cycles.
Electrical measurements		See table V, steps 1, 2, and 4.

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

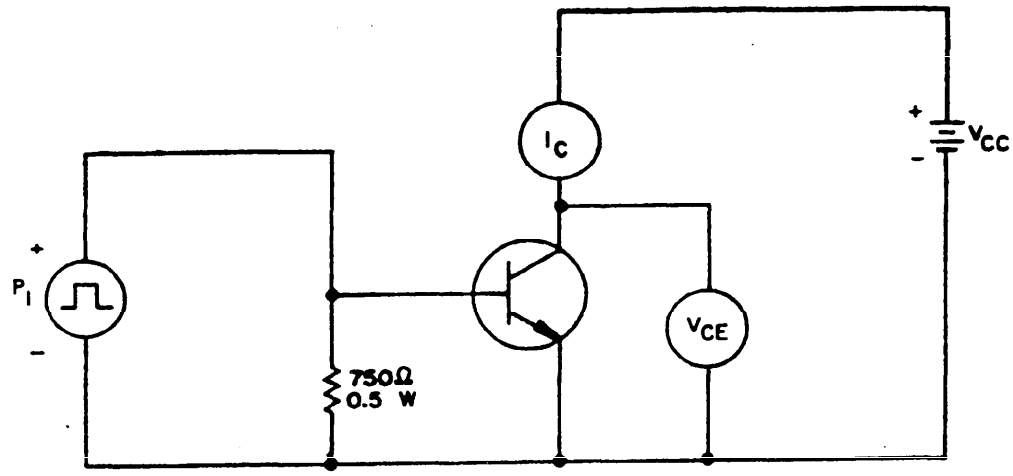
Inspection	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroup 1			22 devices, c = 1
Thermal shock (temperature cycling)	1051	500 cycles.	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7.	
Subgroup 2			22 devices, c = 1
Steady-state dc Blocking life	1039 or 1049	Condition A; 500 hours	
Electrical measurements		See table V, steps 1, 5, 6, and 7.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices, c = 1
Thermal resistance	3131	$R_{\theta JC}$ = 5°C/W maximum. See 4.5.3.	
Subgroup 5			15 devices, c = 1
Barometric pressure (reduced)	1001	Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 s (minimum)	

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TABLE V. Groups A, B, C, and E electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N3584 2N3585	3041	Bias condition A, V_{BE} = -1.5 V dc, V_{CE} = 300 V dc, V_{CE} = 400 V dc	I_{CEX1}		 1.0 1.0	 mA dc mA dc
2.	Breakdown voltage collector to emitter 2N3584 2N3585		I_C = 200 mA dc, See figure 3 and and 4.5.6. Bias condition D	V_{CEO}	 240 285		 V dc V dc
3.	Emitter to base cutoff current	3041	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		0.5	mA dc
4.	Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE1}	25	100	
5.	Collector to emitter cutoff current 2N3584 2N3585	3041	Bias condition A, V_{BE} = -1.5 V dc, V_{CE} = 300 V dc, V_{CE} = 400 V dc	ΔI_{CEX1}	100 percent of initial value or 0.1 mA dc, whichever is greater.		
6.	Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 A dc, pulsed (see 4.5.1)	Δh_{FE1}	±25 percent change in initial recorded value.		
7.	Thermal resistance	3131	See 4.5.2	$R_{\theta JC}$		5.0	°C/W

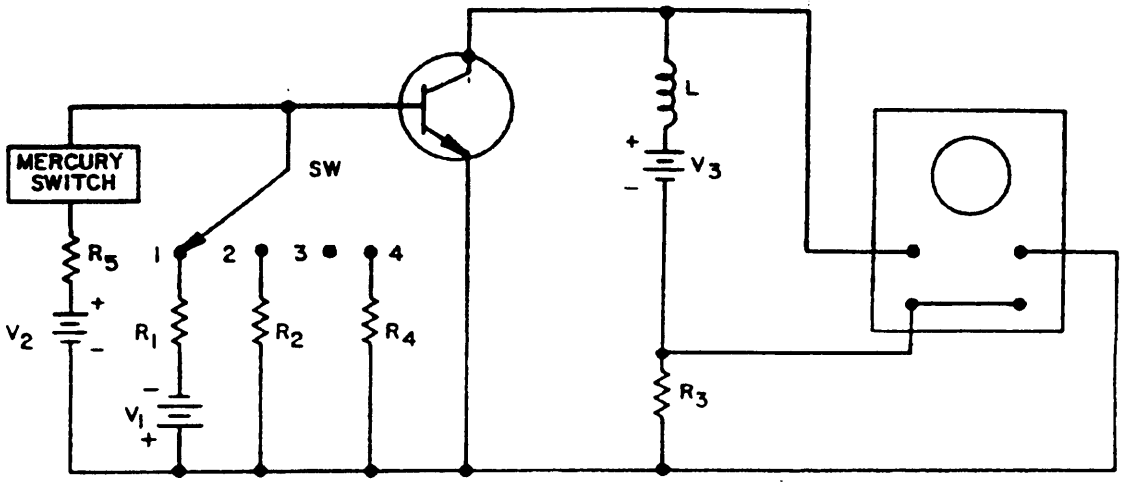
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Adjust V_{CC} and P_1 for V_{CE} = 100 V at I_C = 350 mA.
V_{CC}: 110 V, 1 amp capability, current limited, voltage regulated supply, ±1 percent line and load.
P_1: Pulse current generator, 0 to +140 mA capability.
t_p = 1 second, Test cycles = 1, T_C = +25°C.
I_C: Current indicator (meter or sampling resistor and oscilloscope) with voltage drop less than 1 V at 350 mA.
V_{CE}: Voltage indicator with current requirement of < 1 mA at 100 V (oscilloscope, VTVM or high impedance multimeter)
Reject criteria: Reject if second breakdown occurs, as indicated by collector current sudden increase and becoming uncontrollable by P_1.

FIGURE 2. SBI_C test circuit.

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Mercury switch: Clare mercury relay, model number HGP - 1004, or equivalent.
Oscilloscope: Hewlett Packard oscilloscope, model number 130B, or equivalent

Switch position	Test	L	Reject criteria
1.	SBI_{CEX}	100 µH	SB occurs
2.	SBI_{CER}	500 µH	SB occurs
3.	BV_{CEO}	20 mH	BV_{CEO} < limit
4.	BV_{CER}	5 mH	BV_{CER} < limit

R_1 = 200Ω, 0.5 W
R_2 = 500Ω, 0.5 W
R_3 = 1Ω, 0.5 W
R_4 = 2000Ω, 0.5 W
R_5 = 27Ω, 10 W

V_1 = -4 V
V_2 = 15 V
V_3 = 0 to 50 V, 3 A

FIGURE 3. SBI_{CEX}, SBI_{CER}, BV_{CEO}, BV_{CER} test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Lead material and finish may be specified (see 3.3.1).

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military Part or Identifying Number (PIN). The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

6.3.1 JANTX substitution. JANTX devices are a direct replacement for JAN marked devices (Example: Substitute JANTX2N3584 for JAN2N3584).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 19, 80, 85
DLA - ES
NASA - NA

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15

Preparing activity:
Navy - EC

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/384C	2. DOCUMENT DATE (YYMMDD) 920103

3. DOCUMENT TITLE
Semiconductor device, Transistor, NPN, high-power 2N3584, 2N3585, JANTX and JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Mr. Alan Barone	(513)296-6048	986-6064

c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
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